

COLLEGE OF DEFENCE MANAGEMENT · DATATHON - 2026 · THEME 6.1

PROJECT HIGH GROUND

Technical Annex

Methods, formulas, parameters, validation and limitations behind the Commander's Edition. For assessors and analysts.

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1. SCOPE, DATA AND PREPARATION

This annex documents every method behind the Commander's Edition: definitions, formulas, parameters, validation and limitations. All numbers are regenerated from raw data by *analysis/run_all.py* (12 stages); two consecutive runs give identical outputs.

Table 1.1 : Data sources

Dataset	Size	Period	Fields used	Source
CDM satellite census	1,167 active satellites	launched 1974 - Jan 2014	orbit, mass, power, users, purpose, life	CDM
CDM Dst index	139,872 hourly values	3 anonymised periods	geomagnetic disturbance (nT)	CDM
CDM smoothed sunspot number	192 values	same periods	solar activity	CDM
CDM MODIS fire detections	1,248,605	Sep 2010 - Sep 2020	position, time, satellite, type	CDM
CelesTrak SATCAT	70,366 Earth-orbiting objects	1957 - 21 Sep 2026	type, owner, launch, decay, orbit	Open
UCS Satellite Database	2,666 active satellites	1 Apr 2020	as CDM census (second census)	Open
UNOOSA registry (OWID)	objects per country-year	1957 - 2023	cross-check of launch counts	Open
CelesTrak GP elements (OMM)	15,656 + 4,640 LEO objects	epochs 30 Aug - 23 Sep 2026	SGP4 orbital elements	Open

Table 1.2 : Cleaning log (every step and the rows it touched)

Dataset	Step	Count
CDM satellites	Rows loaded	1,167
CDM satellites	Launch dates converted from Excel serial numbers (1 missing)	1
CDM satellites	Impossible orbital periods (< 80 min) set to missing	6
CDM satellites	Impossible inclinations (> 180 deg) set to missing	1
CDM satellites	Rows after cleaning	1,167
UCS 2020	Active satellites	2,666
SATCAT	Objects catalogued (1957 - Sep 2026)	70,793
SATCAT	Earth-orbiting objects kept	70,366
SATCAT	Objects without orbital elements (kept; excluded from altitude work)	1,657
Space weather	Hourly Dst values (3 periods)	139,872
Space weather	Monthly smoothed sunspot values	192
MODIS fires	Detections loaded (archive + near real time)	1,248,606
MODIS fires	Duplicate detections removed	1
MODIS fires	Rows after cleaning	1,248,605

Validation rules: launch dates converted from Excel serial days (origin 1899-12-30); orbital periods below 80 minutes and inclinations above 180 degrees are physically impossible and set to missing; the misspelt user class 'Commerical' is corrected; catalogue owner and launch-site codes are decoded with explicit look-up tables (analysis/common.py); only Earth-orbiting objects are kept.

2. GROWTH, CONCENTRATION AND CONGESTION

2.1 Population reconstruction

An object is in orbit at date t if launched on or before t and not re-entered by t :

$$N(t) = \#\{ i : \text{launch}_i \leq t < \text{decay}_i \} \quad (\text{decay}_i = \infty \text{ if still in orbit}) \quad (1)$$

Active-satellite counts come from three independent censuses (CDM 2014: 1,167; UCS 2020: 2,666; catalogue operational flag 2026: 17,062). UNOOSA registrations and catalogue payload launches correlate at $r = 0.997$.

2.2 Change-points and growth rates

PELT (Killick et al., 2012) minimises the total within-segment squared error plus a penalty per change-point:

$$\min_{\tau} \sum_k \sum_{t \in \text{segment } k} (y_t - \bar{y}_k)^2 + \beta \cdot K \quad (2)$$

applied to payloads launched per year 1990-2025, scaled by its standard deviation, with $\beta = 3$, minimum segment 3 years. Result: break in 2021. Compound annual growth rate CAGR = $(x_b/x_a)^{1/(b-a)} - 1$: 4.1% (2000-2013), 46.7% (2019-2025).

2.3 Concentration

$$HHI = \sum_c (100 \cdot s_c)^2, \quad s_c = \text{share of active satellites of country } c \quad (3)$$

Thresholds per the US merger guidelines: $> 1,800$ moderately and $> 2,500$ highly concentrated (older convention). Results: 2,171 (2014), 2,780 (2020), 5,879 (2026).

2.4 Collision-risk index

Low orbit (apogee $< 2,000$ km) is cut into 25 km shells at each object's mean altitude (perigee + apogee)/2. Under the kinetic-gas model (Kessler and Cour-Palais, 1978) the collision rate in a shell is proportional to $n_i^2 \cdot \sigma \cdot v_{\text{rel}} / V_i$; with cross-section σ and relative speed v_{rel} held equal across shells:

$$I = \sum_i n_i^2 / V_i, \quad V_i = 4/3 \pi [(R+h_i+\Delta)^3 - (R+h_i)^3], \quad R = 6,371 \text{ km}, \Delta = 25 \text{ km} \quad (4)$$

normalised to January 2014 = 1. Result: 10.3 (Sep 2026); scenarios for 2030: 11.6 / 16.8 / 27.2. It is a relative index, not a probability; it ignores size differences and manoeuvres (which reduce real risk for active satellites) and eccentric orbits crossing several shells.

2.5 Debris survival

For each weapon test or collision, the share of catalogued pieces still in orbit t years after the event, with pieces still in orbit treated as censored at the catalogue date:

$$S(t) = \#\{ \text{pieces with } (\text{decay} - \text{event}) > t \text{ or still in orbit} \} / \#\{ \text{pieces} \} \quad (5)$$

Table 2.1 : Debris created and remaining by event

Event	Date	Pieces catalogued	Still in orbit (Sep 2026)	Still in orbit %	Years since	Median years to re-entry (re-entered pieces)
China ASAT test	Jan 2007	3,531	2,313	65.50	19.70	9.80
Iridium-Cosmos collision (Russian debris)	Feb 2009	1,714	613	35.80	17.60	6.80
Iridium-Cosmos collision (US debris)	Feb 2009	656	117	17.80	17.60	6.90
US ASAT intercept	Feb 2008	174	0	0	18.60	0.10
India ASAT test (Mission Shakti)	Mar 2019	129	0	0	7.50	0.40
Russia ASAT test	Nov 2021	1,806	4	0.20	4.80	0.50

3. CONTEST, SPACE WEATHER AND EARTH-OBSERVATION VALUE

3.1 Programme families (name rules)

Table 3.1 : Military and state-ISR families from official names (lower bound: covert or commercially flagged military satellites are not counted)

Country	Programme family	Operational Sep 2026	Launched since 2014
China	China: Yaogan military reconnaissance	164	149
China	China: other state ISR & tech (Gaofen, TJS, Shiyang, Shijian, Tianhui, Yunhai)	183	173
China	China: commercial imagers (Jilin-1, Superview)	59	94
USA	USA: national-security payloads (USA-nnn)	417	385
USA	USA: commercial imagers (Planet, Maxar, BlackSky, Capella)	125	581
Russia	Russia: military (Cosmos)	146	125
India	India: ISR & military (RISAT, Cartosat, EOS, EMISAT, GSAT-7)	17	15

3.2 Geomagnetic storms

Storm = a run of consecutive hours with $Dst \leq -50$ nT; intense if its minimum ≤ -100 nT, severe if ≤ -200 nT (Gonzalez et al., 1994). Each hour is matched to the smoothed sunspot number by linear interpolation; data are cut into 30-day blocks (blocks with < 25 days dropped). Poisson regression of intense-storm counts:

$$\log E[\text{storms}_b] = \beta_0 + \beta_1 \cdot SSN_b/50 \quad (6)$$

gives a rate ratio $e^{\beta_1} = 1.88$ per 50 sunspots ($p = 1.1e-08$); Spearman ρ between block sunspot number and block minimum $Dst = -0.57$ ($p = 4.8e-18$). Share of 30-day blocks with at least one intense storm: 39% at $SSN \geq 100$ vs 6% at $SSN < 30$.

Table 3.2 : Storm events by class

Class	Events
Moderate (-50 to -100 nT)	498
Intense (-100 to -200 nT)	46
Severe (≤ -200 nT)	13

3.3 Solar cycle in re-entries

Re-entry rate r_y = debris and rocket bodies (perigee $< 2,000$ km at the start of year y) re-entering in year y / that population, 1965-2025. $\log r_y$ is de-trended with a quadratic and a periodogram taken; the dominant period is 12.2 years (solar cycle ~ 11 years). Mean rate near solar maxima 3.9% vs minima 2.3%.

3.4 Constellation premium and surge detection (MODIS)

Each archive detection is assigned to a cell-day (0.1° latitude $\times$ 0.1° longitude $\times$ calendar day). The share of cell-days seen by only one satellite is 84% (Aqua only 46%, Terra only 38%). Monthly surge score $z_m = (x_m - \text{median of the same month in earlier years}) / \text{their standard deviation}$; $z > 2$ flags a surge (11 months).

4. MACHINE LEARNING: MILITARY-USE CLASSIFIER

Table 4.1 : Features

Group	Features
Orbit	log perigee, log apogee, eccentricity, inclination, log period; orbit class (LEO/MEO/GEO/elliptical)
Engineered	sun-synchronous-like (95-100.5 deg, apogee $< 1,500$ km); Molniya-like (eccentricity > 0.5 , 60-66 deg)
Build	log launch mass, log power, design life (gaps allowed: imputed with medians plus missing-value indicators)
Excluded	name, purpose, users, operator, contractor (would leak the label); launch year (cannot extrapolate)
Variant	physics + operator country bloc (USA / China / Russia / India / other)

Label: military = the Users field mentions Military (incl. dual-use), 332 of 1,167 in 2014. Models: L2-regularised logistic regression ($C = 0.5$) and histogram gradient-boosted trees (250 iterations, learning rate 0.05, 15 leaves). Protocol: (a) 5-fold stratified cross-validation repeated 5 times on the 2014 census; (b) out-of-time test: fit on all of 2014, predict the 1,851 UCS 2020 satellites launched after the census (238 military). The model is chosen on (b).

Table 4.2 : Model skill

Features	Model	CV ROC-AUC 2014 (mean)	CV SD	Out-of-time ROC-AUC (2020 launches)	Out-of-time PR-AUC	Balanced accuracy @0.5
Physics only	Logistic regression	0.83	0.00	0.79	0.53	0.70
Physics only	Gradient-boosted trees	0.94	0.00	0.64	0.27	0.62
Physics + operator country	Logistic regression	0.84	0.00	0.76	0.55	0.71
Physics + operator country	Gradient-boosted trees	0.94	0.00	0.63	0.27	0.61

Table 4.3 : Permutation importance on the out-of-time set (chosen model)

Feature	AUC drop when shuffled
Launch mass	0.16
Apogee	0.11
Sun-synchronous-like	0.05
Design life	0.04
MEO	0.02
Inclination	0.01
Molniya-like	0.01
Period	0.00

Cross-validation would have chosen the trees (0.941), which collapse out of time (0.644): the post-2014 population (mass-produced commercial constellations in low orbit) differs from the 2014 one, and the trees memorise the old one. The logistic model degrades gracefully (0.835 to 0.793). Dual-use signal: 83 of 1,613 non-military satellites score $p \geq 0.5$.

5. CLOSE-APPROACH SCREENING

Table 5.1 : Screening parameters

Item	Setting
Elements	CelesTrak GP data (CCSDS OMM) via public mirror; LEO only: mean motion ≥ 11.25 rev/day, eccentricity ≤ 0.25
Screening A	15,656 active satellites vs each other, 24 h from 31 Aug 2026 12:00 UTC (elements of 30-31 Aug)
Screening B	4,640 objects: freshly tracked satellites + 4 debris clouds, 24 h from 23 Sep 2026 12:00 UTC (elements of 21-24 Sep)
Propagator	SGP4 (Vallado et al., 2006), WGS-72, vectorised
Coarse filter	positions every 10 s; k-d tree pairs within 80 km ($\geq$ half the 150 km two objects can close in 10 s at 15 km/s)
Refinement	1-s positions over ± 10 s around the closest sample; linearised closest approach (eq. 7)
Thresholds	report < 5 km; high interest < 1 km; co-orbiting / docked if relative speed < 0.5 km/s

$$\tau^* = -(\Delta r \cdot \Delta v) / |\Delta v|^2, \quad \text{miss} = |\Delta r + \Delta v \tau^*| \quad (\tau^* \text{ clipped to } \pm 1 \text{ s}) \quad (7)$$

Table 5.2 : Encounters under 5 km in the two 24-hour screenings

screen	kind	encounters_5km	under_1km	median_rel_speed
A: active vs active (31 Aug-1 Sep)	Co-orbiting (formation or docked)	430	86	0.15
A: active vs active (31 Aug-1 Sep)	Different operators, crossing	5,043	193	11.18
A: active vs active (31 Aug-1 Sep)	Same constellation, crossing	12,844	192	9.78
B: satellites vs debris clouds (23-24 Sep)	Satellite vs debris	468	20	13.09

Table 5.3 : Closest crossing encounters

Object A	Object B	Kind	Miss (km)	Speed (km/s)	Altitude (km)
YAOGAN-14	STARLINK-5061	Different operators, crossing	0.05	14.34	482
STARLINK-5366	STARLINK-36147	Same constellation, crossing	0.06	3.89	485
STARLINK-30385	STARLINK-32434	Same constellation, crossing	0.11	7.91	485
STARLINK-31971	STARLINK-35986	Same constellation, crossing	0.11	1.26	477
STARLINK-3122	STARLINK-4550	Same constellation, crossing	0.11	12.16	465
KSLV-II DUMMY	GAOFEN-5 01A	Different operators, crossing	0.12	14.57	693
JILIN-1 GAOFEN 03D48	STARLINK-37742	Different operators, crossing	0.13	14.74	461
STARLINK-6327	STARLINK-36432	Same constellation, crossing	0.13	1.28	485
AL MUNTER	STARLINK-35382	Different operators, crossing	0.13	13.71	467
YAOGAN-12	STARLINK-6173	Different operators, crossing	0.14	6.98	483

Limits: public GP elements carry about 1 km error, growing with time from epoch, so miss distances are screening values; no covariance or collision probability is computed; objects manoeuvre; the full debris catalogue (about 12,500 pieces) is not publicly mirrored, so screening B covers the four largest clouds only; the catalogue excludes objects smaller than about 10 cm.

6. FORECAST, LIMITATIONS AND REPRODUCIBILITY

6.1 Forecast

Holt's damped additive trend on log payloads in orbit, 2000-2026 (Hyndman and Athanasopoulos, 2021):

$$\ell_t = \alpha y_t + (1-\alpha)(\ell_{t-1} + \phi b_{t-1}), \quad b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)\phi b_{t-1} \quad (8)$$

80% interval from 2,000 simulated paths (seeded). Back-test: fit to 2000-2021, forecast 2022-2026, mean absolute percentage error 8.6%. 2030: 36,000 (28,000-46,100).

Table 6.1 : Bottom-up scenarios for working satellites in 2030 (others grow 3 / 7 / 12% a year)

Scenario	Starlink (US)	Kuiper (US)	Guowang (China)	Qianfan (China)	All other satellites	Total 2030
Low	12,000	1,600	800	800	5,701	20,901
Base	15,000	3,236	3,000	3,000	6,639	30,875
High	19,400	3,236	6,500	6,500	7,970	43,606

6.2 Limitations and threats to validity

- The CDM census is a 2014 snapshot; growth after 2014 rests on the open catalogue and the UCS 2020 census (cross-checked).
- 'Operational' in the catalogue is the maintainer's flag; some dead satellites may be flagged operational and vice versa.
- Military counts use names: a lower bound. The classifier's label is the registered user, itself imperfect for dual-use satellites.
- The collision-risk index is relative and ignores object size, manoeuvres and eccentric orbits spanning shells.
- The CDM space-weather periods are anonymised; the storm-sunspot relation uses their internal timing only.

- Scenario totals depend on announced plans; they are assumptions, stated in the table, not predictions of company behaviour.

6.3 Reproducibility

Python 3.11.15; pandas 3.0.6; NumPy 2.3.5; SciPy 1.17.1; statsmodels 0.15.0; scikit-learn 1.9.1; ruptures 1.1.10; sgp4 2.27; Matplotlib 3.10.9. Commands: *cd analysis && python run_all.py* (about 8 minutes, of which 7 are the close-approach screening), then *cd ../report && python build_booklet.py && python build_annex.py && python build_word.py*. Random elements are seeded; two consecutive runs give identical metrics.